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Remote CARI (rCARI) Validation Study

CAR Report 2.0

SAVING LIVES

CHANGING LIVES

RAM-N – rCARI Team | May 2024

Background

- Objective is to validate the remote Consolidated Approach for Reporting Indicators of Food Insecurity (rCARI)
- Current report is for Central African Republic
  - Data collection was July-August 2023

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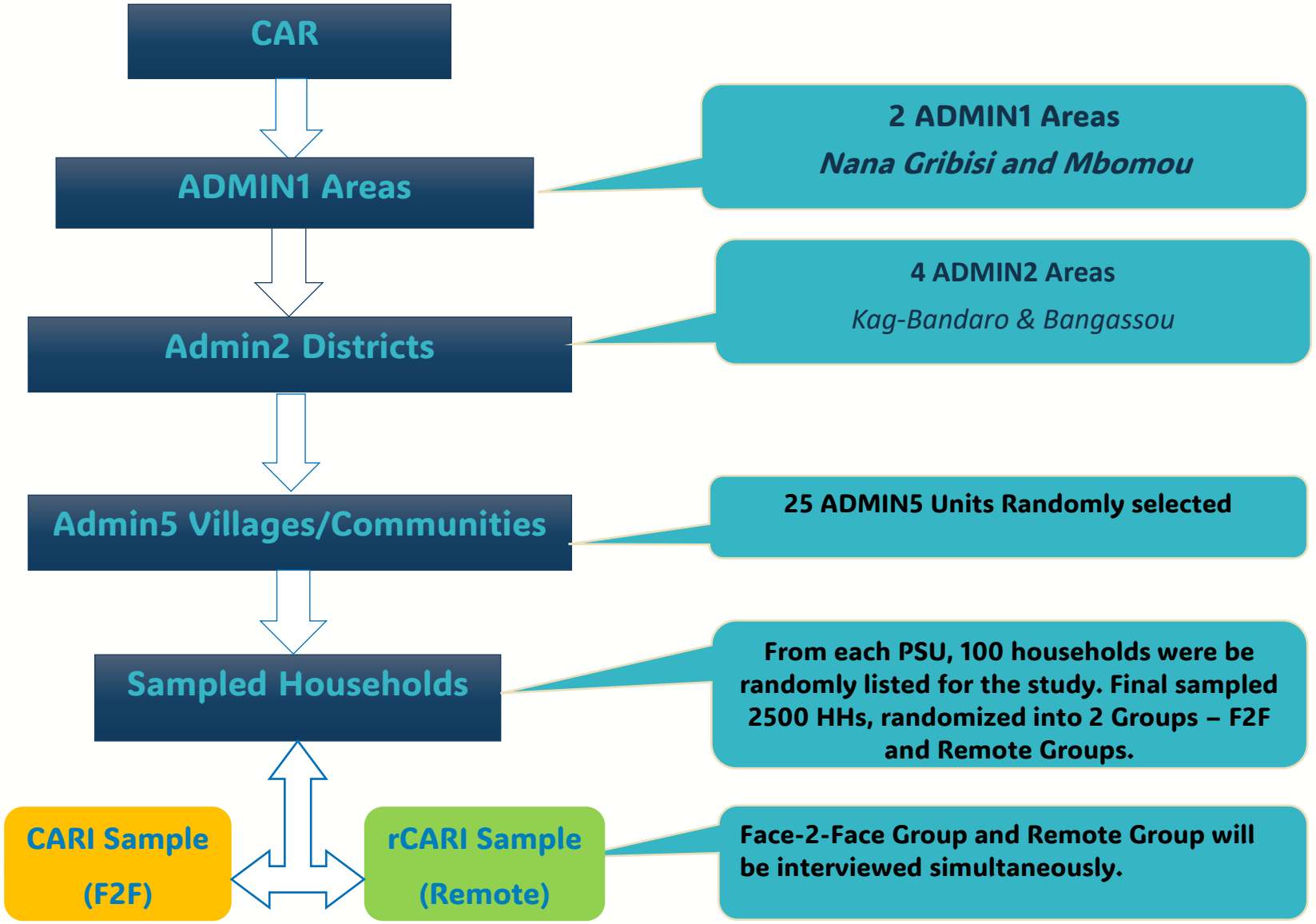
Implementation



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Study design considerations

- Mobile phone penetration
    - Mobile phone network
- Accessibility
- AFI

These factors together informed the final ADMIN selection

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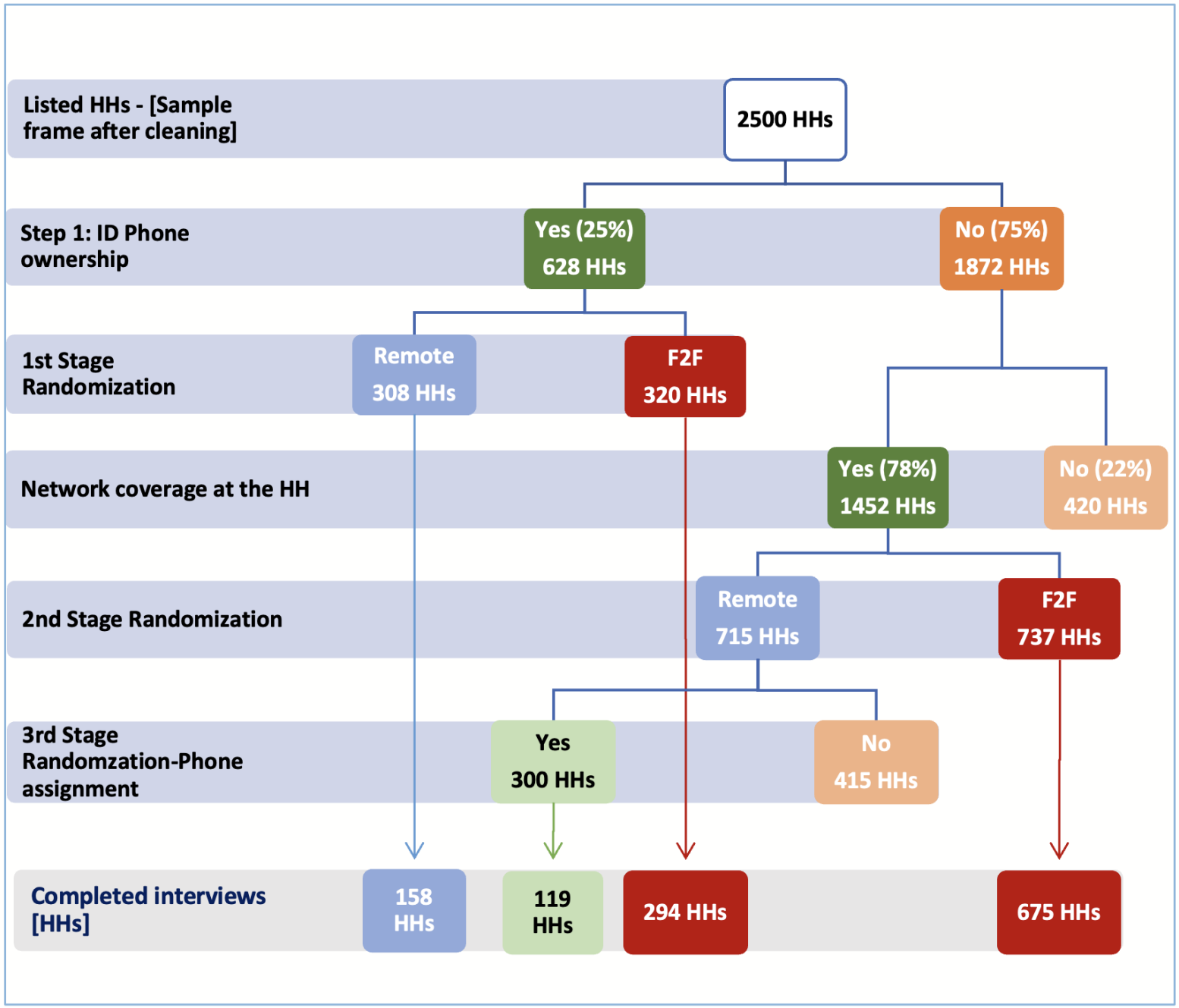
Sampling & completion



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Distribution of ADMIN5s

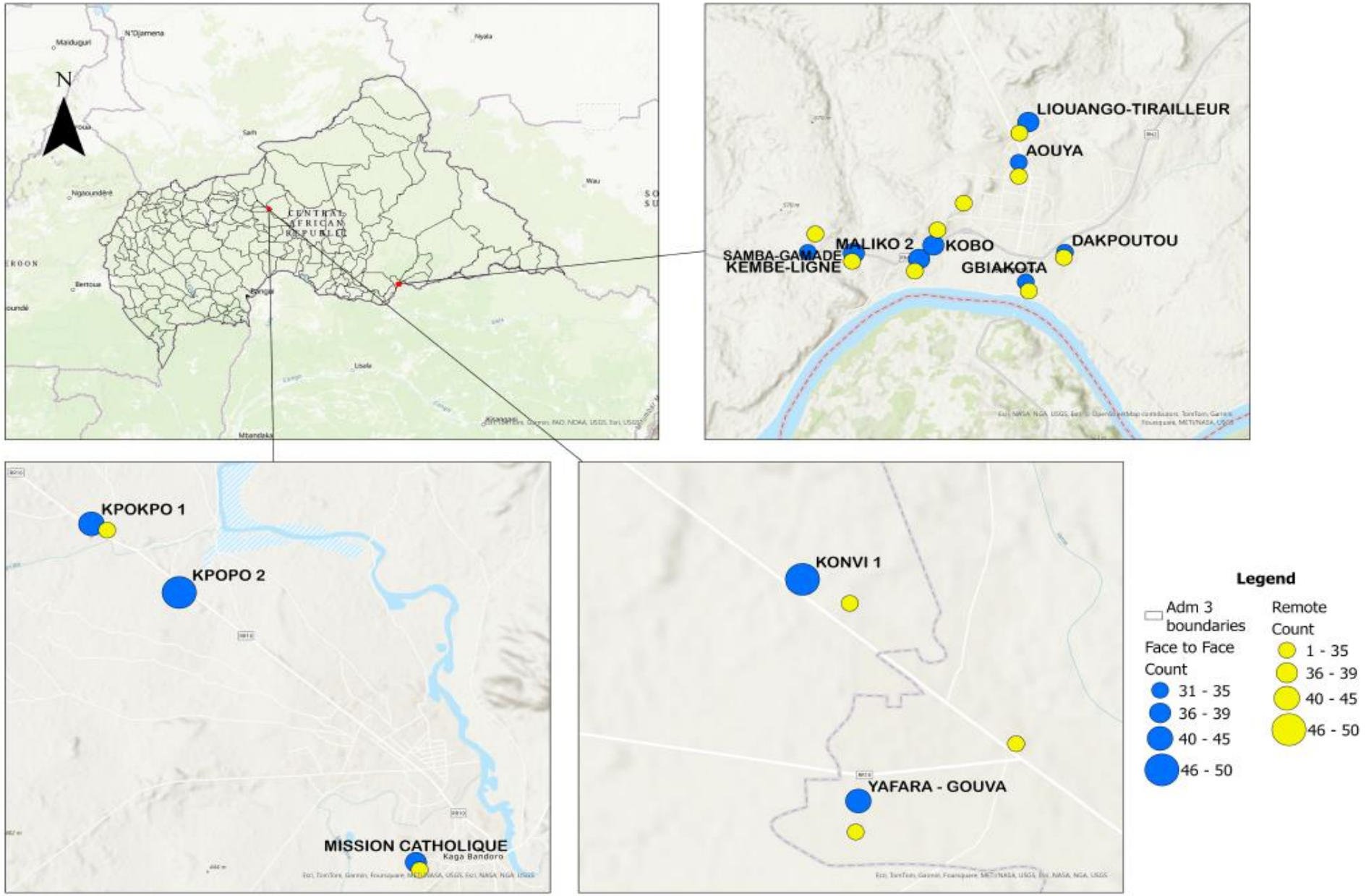

Figure 3: Distributions of completed interviews in the study area

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Balance tests for randomization

Balance Table for Randomized Survey Modality Assignment: Face-to-Face vs. Remote

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
List respondent age	35.97	37.59	1.61	39.42	39.35	-0.08	38.38	38.82	0.44
	(12.42)	(12.76)	(0.11)	(13.95)	(14.24)	(0.92)	(13.59)	(13.83)	(0.47)
Household head is the respondent	0.72	0.73	0.01	0.78	0.79	0.02	0.76	0.78	0.02
	(0.45)	(0.44)	(0.68)	(0.42)	(0.40)	(0.43)	(0.43)	(0.42)	(0.38)
Female Household Head	0.20	0.23	0.03	0.30	0.31	0.01	0.27	0.28	0.02
	(0.40)	(0.42)	(0.30)	(0.46)	(0.46)	(0.66)	(0.44)	(0.45)	(0.37)
Married Household Head	0.25	0.26	0.01	0.14	0.11	-0.03	0.17	0.16	-0.02
	(0.44)	(0.44)	(0.85)	(0.35)	(0.31)	(0.09)	(0.38)	(0.36)	(0.25)
Household head years of education	7.50	7.49	-0.02	3.39	3.30	-0.08	4.63	4.56	-0.07
	(4.10)	(4.50)	(0.96)	(3.27)	(3.22)	(0.63)	(4.01)	(4.12)	(0.70)
Displaced household	0.11	0.12	0.01	0.03	0.03	-0.00	0.06	0.06	0.00
	(0.31)	(0.32)	(0.67)	(0.18)	(0.18)	(0.85)	(0.23)	(0.23)	(0.86)
Household size	7.68	7.95	0.28	6.09	5.96	-0.13	6.57	6.56	-0.01
	(3.50)	(3.95)	(0.35)	(2.66)	(2.65)	(0.37)	(3.03)	(3.23)	(0.96)
Number of household member younger than 15	4.02	4.00	-0.02	3.21	3.13	-0.08	3.45	3.39	-0.06
	(2.27)	(2.12)	(0.93)	(1.92)	(1.90)	(0.41)	(2.06)	(2.01)	(0.47)
Number of household member older than 60	0.21	0.22	0.01	0.18	0.18	0.00	0.19	0.19	0.01
	(0.48)	(0.55)	(0.72)	(0.44)	(0.48)	(0.87)	(0.45)	(0.50)	(0.73)
Has household member with disability/chronic disease	0.29	0.30	0.00	0.15	0.16	0.01	0.19	0.20	0.01
	(0.46)	(0.46)	(0.96)	(0.36)	(0.37)	(0.50)	(0.40)	(0.40)	(0.59)
Number of rooms	2.84	2.99	0.15	1.96	1.91	-0.05	2.23	2.24	0.01
	(1.25)	(1.34)	(0.14)	(1.08)	(1.09)	(0.40)	(1.20)	(1.27)	(0.83)
Household Crowding Index	3.08	3.00	-0.08	4.02	4.00	-0.02	3.74	3.70	-0.03
	(1.79)	(1.64)	(0.57)	(2.94)	(2.97)	(0.92)	(2.68)	(2.68)	(0.78)
Household has impoverished roof	0.67	0.66	-0.01	0.85	0.86	0.01	0.79	0.80	0.00
	(0.47)	(0.48)	(0.80)	(0.36)	(0.35)	(0.57)	(0.41)	(0.40)	(0.79)
Household has impoverished floor	0.89	0.91	0.02	0.96	0.96	0.00	0.94	0.95	0.01
	(0.32)	(0.29)	(0.37)	(0.19)	(0.19)	(0.87)	(0.24)	(0.22)	(0.44)
Observations	320	308	628	737	715	1,452	1,057	1,023	2,080

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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Survey completion

Table 1: Sample Completion: Face-to-Face vs. Remote

	Not Completed		Completed		Total	
	Count	Percent	Count	Percent	Count	Percent
Face-to-Face	88	8.33	969	91.67	1057	100.00
Remote	747	73.02	276	26.98	1023	100.00
Total	835	40.14	1245	59.86	2080	100.00

SHARE ON

Balance tests for phone sample

Table 2: Balance Table by Randomized Survey Modality: Face-to-Face vs. Remote for Completed Surveys

	Own Phone		
Variable	F2F	Remote	Diff
List respondent age	36.27	39.03	2.76*
	(12.55)	(12.38)	(0.03)
Household head is the respondent	0.72	0.78	0.05
	(0.45)	(0.42)	(0.21)
Female Household Head	0.19	0.25	0.06
	(0.39)	(0.44)	(0.12)
Married Household Head	0.27	0.27	0.01
	(0.44)	(0.45)	(0.85)
Household head years of education	7.55	7.92	0.38
	(4.13)	(4.69)	(0.40)
Displaced household	0.09	0.15	0.06
	(0.29)	(0.36)	(0.07)
Household size	7.72	8.39	0.67*
	(3.44)	(3.40)	(0.05)
Number of household member younger than 15	4.04	4.31	0.27
	(2.27)	(2.03)	(0.20)
Number of household member older than 60	0.21	0.21	-0.00
	(0.49)	(0.44)	(0.99)
Has household member with disability/chronic disease	0.30	0.31	0.01
	(0.46)	(0.46)	(0.83)
Number of rooms	2.89	3.13	0.24
	(1.24)	(1.27)	(0.05)
Household Crowding Index	3.02	2.97	-0.05
	(1.70)	(1.26)	(0.73)
Household has impoverished roof	0.68	0.63	-0.05
	(0.47)	(0.48)	(0.33)
Household has impoverished floor	0.90	0.92	0.02
	(0.30)	(0.28)	(0.50)
Observations	294	157	451

Note: Robust standard errors in parentheses; * * p<0.05, ** p<0.01, *** p<0.001.

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Balance tests for non phone sample

Table 2: Balance Table by Randomized Survey Modality: Face-to-Face vs. Remote for Completed Surveys

	No Phone		
Variable	F2F	Remote	Diff
List respondent age	39.47	39.76	0.28
	(13.82)	(13.94)	(0.84)
Household head is the respondent	0.79	0.82	0.04
	(0.41)	(0.38)	(0.34)
Female Household Head	0.30	0.32	0.02
	(0.46)	(0.47)	(0.62)
Married Household Head	0.14	0.14	-0.00
	(0.35)	(0.35)	(0.98)
Household head years of education	3.41	3.54	0.13
	(3.25)	(3.49)	(0.70)
Displaced household	0.03	0.03	0.00
	(0.17)	(0.18)	(0.89)
Household size	6.08	5.99	-0.08
	(2.67)	(3.05)	(0.78)
Number of household member younger than 15	3.23	3.13	-0.10
	(1.95)	(2.29)	(0.64)
Number of household member older than 60	0.17	0.13	-0.04
	(0.44)	(0.41)	(0.33)
Has household member with disability/chronic diseas	0.15	0.15	0.00
	(0.35)	(0.36)	(0.90)
Number of rooms	1.99	1.95	-0.04
	(1.08)	(1.09)	(0.72)
Household Crowding Index	3.93	3.80	-0.13
	(2.87)	(2.68)	(0.62)
Household has impoverished roof	0.85	0.86	0.01
	(0.36)	(0.35)	(0.78)
Household has impoverished floor	0.96	0.97	0.01
	(0.19)	(0.16)	(0.46)
Observations	675	119	794

Note: Robust standard errors in parentheses; * * p<0.05, ** p<0.01, *** p<0.001.

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CARI & rCARI Aggregation

CARI vs. rCARI

Current Situation

Food Consumption Score

reduced Coping Strategy Index

Coping Capacity

Economic Vulnerability

- **CARI:** ECMEN or FES
- **rCARI:** Income Source and Change

Livelihood-based Coping Strategies for Food Security

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Variations of the Aggregation

Table 3: The different versions of CARI & their indicators

Indicator	Aggregation Methodology			
	CARI (F2F)	CARI (F2F)	rCARI (Remote)	rCARI (Remote)
FCS	Yes	Yes	Yes	Yes
FES	Yes	No	Yes	No
LCS	Yes	Yes	Yes	Yes
rCSI	Yes	Yes	Yes	Yes
Income source & income change	No	Yes	No	Yes

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CARI vs. rCARI Comparison

Table 4: Impact of Survey Modality: Standard CARI-FES vs. Standard rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.01	0.01	-0.00	0.00	0.02	0.02	0.00	0.01	0.01
	(0.08)	(0.08)	(0.96)	(0.00)	(0.13)	(0.15)	(0.05)	(0.10)	(0.17)
Marginally Food Secure	0.31	0.52	0.21***	0.07	0.14	0.07*	0.14	0.36	0.22***
	(0.46)	(0.50)	(0.00)	(0.25)	(0.35)	(0.03)	(0.35)	(0.48)	(0.00)
Moderately Food Insecure	0.60	0.45	-0.15**	0.74	0.70	-0.04	0.70	0.56	-0.14***
	(0.49)	(0.50)	(0.00)	(0.44)	(0.46)	(0.39)	(0.46)	(0.50)	(0.00)
Severely Food Insecure	0.08	0.02	-0.06**	0.20	0.14	-0.05	0.16	0.07	-0.09***
	(0.27)	(0.14)	(0.00)	(0.40)	(0.35)	(0.14)	(0.37)	(0.26)	(0.00)
Food Insecure Group	0.68	0.47	-0.21***	0.93	0.84	-0.09**	0.86	0.63	-0.23***
	(0.47)	(0.50)	(0.00)	(0.25)	(0.37)	(0.01)	(0.35)	(0.48)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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Source: rCARI survey 2023

CARI vs. rCARI Comparison [FES variable]

Table 5: Impact of Survey Modality: CARI-FES vs. rCARI-FES

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.01	0.01	-0.00	0.00	0.01	0.01	0.00	0.01	0.01
	(0.08)	(0.08)	(0.96)	(0.00)	(0.09)	(0.32)	(0.05)	(0.08)	(0.33)
Marginally Food Secure	0.31	0.29	-0.02	0.07	0.13	0.06	0.14	0.22	0.08**
	(0.46)	(0.45)	(0.61)	(0.25)	(0.33)	(0.07)	(0.35)	(0.41)	(0.01)
Moderately Food Insecure	0.60	0.58	-0.02	0.74	0.59	-0.15**	0.70	0.58	-0.11***
	(0.49)	(0.50)	(0.65)	(0.44)	(0.49)	(0.00)	(0.46)	(0.49)	(0.00)
Severely Food Insecure	0.08	0.13	0.05	0.20	0.28	0.08	0.16	0.19	0.03
	(0.27)	(0.33)	(0.14)	(0.40)	(0.45)	(0.06)	(0.37)	(0.39)	(0.24)
Food Insecure Group	0.68	0.71	0.02	0.93	0.87	-0.07*	0.86	0.78	-0.08**
	(0.47)	(0.46)	(0.61)	(0.25)	(0.34)	(0.04)	(0.35)	(0.42)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Source: rCARI survey 2023

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CARI vs. rCARI Comparison [Income variable]

Table 6: Impact of Survey Modality: CARI-Inc vs. rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.02	0.01	-0.01	0.00	0.02	0.02	0.01	0.01	0.00
	(0.14)	(0.08)	(0.18)	(0.04)	(0.13)	(0.20)	(0.08)	(0.10)	(0.59)
Marginally Food Secure	0.51	0.52	0.01	0.24	0.14	-0.09**	0.32	0.36	0.04
	(0.50)	(0.50)	(0.86)	(0.43)	(0.35)	(0.01)	(0.47)	(0.48)	(0.25)
Moderately Food Insecure	0.45	0.45	0.01	0.72	0.70	-0.03	0.64	0.56	-0.08*
	(0.50)	(0.50)	(0.89)	(0.45)	(0.46)	(0.55)	(0.48)	(0.50)	(0.02)
Severely Food Insecure	0.02	0.02	-0.00	0.04	0.14	0.11**	0.03	0.07	0.04*
	(0.14)	(0.14)	(0.92)	(0.19)	(0.35)	(0.00)	(0.18)	(0.26)	(0.01)
Food Insecure Group	0.47	0.47	0.01	0.76	0.84	0.08*	0.67	0.63	-0.04
	(0.50)	(0.50)	(0.91)	(0.43)	(0.37)	(0.04)	(0.47)	(0.48)	(0.21)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Source: rCARI survey 2023

DO NOT INTERNAL

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CARI vs. rCARI Comparison [Paired Responses]

Table 7: Impact of Indicator Composition: CARI-FES vs. CARI-Inc in F2F Surveys (Paired)

	Own Phone			No Phone			Total		
Food Security Category	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff
Food Secure	0.01	0.02	0.01**	0.00	0.00	0.00	0.00	0.01	0.01*
	(0.00)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Marginally Food Secure	0.31	0.51	0.2***	0.07	0.24	0.17***	0.14	0.32	0.18***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)
Moderately Food Insecure	0.60	0.45	-0.16***	0.74	0.72	-0.01	0.70	0.64	-0.06**
	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.01)	(0.02)	(0.02)
Severely Food Insecure	0.08	0.02	-0.06***	0.20	0.04	-0.16***	0.16	0.03	-0.13***
	(0.02)	(0.01)	(0.02)	(0.02)	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)
Food Insecure Group	0.68	0.47	-0.22***	0.93	0.76	-0.17***	0.86	0.67	-0.18***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)
Observations	294	294	294	675	675	675	969	969	969

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

REDACTED

Source: rCARI survey 2023

CARI vs. rCARI Comparison [Paired Responses]

Table 8: Impact of Indicator Composition: CARI-FES vs. CARI-Inc in Remote Surveys (Paired)

	Own Phone			No Phone			Total		
Food Security Category	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff
Food Secure	0.01	0.01	0.00	0.01	0.02	0.01	0.01	0.01	0.00
	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)
Marginally Food Secure	0.29	0.52	0.24***	0.13	0.14	0.02	0.22	0.36	0.14***
	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)
Moderately Food Insecure	0.58	0.45	-0.13*	0.59	0.70	0.11*	0.58	0.56	-0.03
	(0.04)	(0.04)	(0.05)	(0.05)	(0.04)	(0.05)	(0.03)	(0.03)	(0.04)
Severely Food Insecure	0.13	0.02	-0.11***	0.28	0.14	-0.13**	0.19	0.07	-0.12***
	(0.03)	(0.01)	(0.03)	(0.04)	(0.03)	(0.04)	(0.02)	(0.02)	(0.02)
Food Insecure Group	0.71	0.47	-0.24***	0.87	0.84	-0.03	0.78	0.63	-0.14***
	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Observations	157	157	157	119	119	119	276	276	276

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Source: rCARI survey 2023

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Individual Food Security Indicators - FCS

Table 9: Impact of Survey Modality on Food Consumption Score and Groups

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Consumption Score	39.76	47.63	7.87***	24.59	28.12	3.53**	29.19	39.22	10.03***
	(17.58)	(17.45)	(0.00)	(12.67)	(12.00)	(0.00)	(15.94)	(18.12)	(0.00)
FCS: Acceptable	0.40	0.58	0.18***	0.10	0.09	-0.01	0.19	0.37	0.18***
	(0.49)	(0.50)	(0.00)	(0.31)	(0.29)	(0.70)	(0.40)	(0.48)	(0.00)
FCS: Borderline	0.35	0.29	-0.06	0.24	0.39	0.15**	0.27	0.33	0.06
	(0.48)	(0.45)	(0.19)	(0.43)	(0.49)	(0.00)	(0.45)	(0.47)	(0.07)
FCS: Poor	0.25	0.13	-0.12**	0.66	0.52	-0.14**	0.53	0.30	-0.23***
	(0.43)	(0.34)	(0.00)	(0.48)	(0.50)	(0.01)	(0.50)	(0.46)	(0.00)
Consumption over the past 7 days (cereals and tubers)	5.87	6.24	0.36*	5.08	4.87	-0.21	5.32	5.64	0.33*
	(1.86)	(1.33)	(0.02)	(2.17)	(2.13)	(0.31)	(2.11)	(1.85)	(0.01)
Consumption over the past 7 days (pulses)	3.74	3.74	-0.01	2.01	1.98	-0.03	2.54	2.98	0.45**
	(2.33)	(2.08)	(0.98)	(1.92)	(1.62)	(0.88)	(2.20)	(2.08)	(0.00)
Consumption over the past 7 days (dairy products)	0.50	1.14	0.64**	0.07	0.40	0.33**	0.20	0.82	0.62***
	(1.57)	(2.27)	(0.00)	(0.55)	(1.11)	(0.00)	(0.99)	(1.90)	(0.00)
Consumption over the past 7 days (protein-rich foods)	1.72	2.09	0.37*	0.51	0.59	0.08	0.88	1.44	0.56***
	(1.81)	(1.81)	(0.04)	(1.07)	(0.99)	(0.45)	(1.45)	(1.68)	(0.00)
Consumption over the past 7 days (vegetables)	3.73	5.64	1.92***	4.11	5.56	1.46***	3.99	5.61	1.62***
	(2.93)	(1.66)	(0.00)	(2.75)	(1.78)	(0.00)	(2.81)	(1.71)	(0.00)
Consumption over the past 7 days (fruit)	0.54	0.72	0.18	0.31	0.59	0.27*	0.38	0.66	0.28**
	(1.31)	(1.34)	(0.17)	(1.03)	(1.22)	(0.02)	(1.13)	(1.29)	(0.00)
Consumption over the past 7 days (oil)	4.90	5.79	0.89***	1.99	2.78	0.79***	2.87	4.49	1.62***
	(2.37)	(1.81)	(0.00)	(2.38)	(2.24)	(0.00)	(2.73)	(2.50)	(0.00)
Consumption over the past 7 days (sugar)	2.36	3.53	1.17***	1.30	1.86	0.55**	1.62	2.81	1.18***
	(2.60)	(2.78)	(0.00)	(2.02)	(1.98)	(0.01)	(2.26)	(2.60)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Source: rCARI survey 2023

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Individual Food Security Indicators - rCSI

Table 10: Impact of Survey Modality on Reduced Coping Strategy Index Classification

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Reduced Coping Strategies Index (rCSI)	14.35	17.27	2.92**	13.16	16.75	3.59***	13.52	17.05	3.52***
	(9.97)	(8.88)	(0.00)	(10.70)	(9.82)	(0.00)	(10.49)	(9.28)	(0.00)
rCSI: Low	0.06	0.04	-0.02	0.19	0.07	-0.13***	0.15	0.05	-0.10***
	(0.25)	(0.21)	(0.36)	(0.40)	(0.25)	(0.00)	(0.36)	(0.23)	(0.00)
rCSI: Median	0.70	0.57	-0.14**	0.55	0.57	0.02	0.60	0.57	-0.03
	(0.46)	(0.50)	(0.00)	(0.50)	(0.50)	(0.66)	(0.49)	(0.50)	(0.41)
rCSI: Severe	0.23	0.39	0.16***	0.26	0.36	0.11*	0.25	0.38	0.13***
	(0.42)	(0.49)	(0.00)	(0.44)	(0.48)	(0.03)	(0.43)	(0.49)	(0.00)
Relied on less preferred, less expensive food	3.92	2.75	-1.17***	3.39	3.29	-0.10	3.55	2.98	-0.57***
	(2.00)	(1.84)	(0.00)	(2.43)	(2.45)	(0.67)	(2.32)	(2.14)	(0.00)
Borrowed food or relied on help from friends or relatives	0.77	0.89	0.12	0.82	0.74	-0.08	0.81	0.82	0.02
	(1.15)	(1.25)	(0.32)	(1.26)	(1.03)	(0.43)	(1.23)	(1.16)	(0.84)
Reduced portion size of meals at meals time	1.98	2.65	0.67**	2.01	2.45	0.44*	2.00	2.57	0.57***
	(2.27)	(2.14)	(0.00)	(2.39)	(2.09)	(0.04)	(2.35)	(2.12)	(0.00)
Restricted consumption by adults in order for young-children to eat	1.73	2.70	0.97***	1.52	2.42	0.90***	1.58	2.58	1.00***
	(1.90)	(1.83)	(0.00)	(1.80)	(2.16)	(0.00)	(1.83)	(1.98)	(0.00)
Reduced the number of meals eaten per day	1.74	2.00	0.26	1.55	2.27	0.72***	1.61	2.12	0.51***
	(2.38)	(2.12)	(0.24)	(2.42)	(2.06)	(0.00)	(2.41)	(2.09)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Source: rCARI survey 2023

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Individual Food Security Indicators - LCS

Table 11: Impact of Survey Modality on Livelihood Coping Strategy

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Maximum Livelihood Coping: None	0.20	0.11	-0.09*	0.39	0.13	-0.26***	0.33	0.12	-0.21***
	(0.40)	(0.32)	(0.01)	(0.49)	(0.34)	(0.00)	(0.47)	(0.33)	(0.00)
Maximum Livelihood Coping: Stress level	0.61	0.43	-0.19***	0.43	0.39	-0.04	0.48	0.41	-0.07*
	(0.49)	(0.50)	(0.00)	(0.50)	(0.49)	(0.39)	(0.50)	(0.49)	(0.03)
Maximum Livelihood Coping: Crisis level	0.14	0.26	0.13**	0.13	0.21	0.08*	0.13	0.24	0.11***
	(0.34)	(0.44)	(0.00)	(0.33)	(0.41)	(0.04)	(0.34)	(0.43)	(0.00)
Maximum Livelihood Coping: Emergency level	0.05	0.20	0.15***	0.05	0.27	0.22***	0.05	0.23	0.18***
	(0.22)	(0.40)	(0.00)	(0.22)	(0.45)	(0.00)	(0.22)	(0.42)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * $p<0.05$, ** $p<0.01$, *** $p<0.001$.

Source: rCARI survey 2023

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Individual Food Security Indicators - FES

Table 13: Impact of Survey Modality on Expenditure and Food Expenditure Share

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Expenditure Share	0.67	0.73	0.06*	0.73	0.68	-0.06*	0.71	0.70	-0.01
	(0.20)	(0.24)	(0.01)	(0.17)	(0.25)	(0.02)	(0.18)	(0.24)	(0.52)
FES < 50 percent	0.20	0.16	-0.04	0.10	0.25	0.15***	0.13	0.20	0.07*
	(0.40)	(0.37)	(0.27)	(0.30)	(0.44)	(0.00)	(0.34)	(0.40)	(0.01)
FES: 50-65 percent	0.23	0.17	-0.06	0.16	0.18	0.02	0.18	0.17	-0.00
	(0.42)	(0.38)	(0.15)	(0.36)	(0.38)	(0.61)	(0.38)	(0.38)	(0.86)
FES: 65-75 percent	0.13	0.13	-0.01	0.17	0.11	-0.06	0.16	0.12	-0.04
	(0.34)	(0.33)	(0.87)	(0.38)	(0.31)	(0.06)	(0.37)	(0.33)	(0.08)
FES: >= 75 percent	0.44	0.54	0.10*	0.57	0.46	-0.11*	0.53	0.51	-0.02
	(0.50)	(0.50)	(0.04)	(0.50)	(0.50)	(0.03)	(0.50)	(0.50)	(0.50)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Source: rCARI survey 2023

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Individual Food Security Indicators - Inc

Table 14: Impact of Survey Modality on Income

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Income: Regular and Unchanged/Increasing	0.45	0.26	-0.19***	0.44	0.24	-0.19***	0.44	0.25	-0.19***
	(0.50)	(0.44)	(0.00)	(0.50)	(0.43)	(0.00)	(0.50)	(0.44)	(0.00)
Income: Regular and Reduced OR Irregular and Unchanged/Increasing	0.42	0.50	0.08	0.46	0.49	0.02	0.45	0.49	0.04
	(0.49)	(0.50)	(0.13)	(0.50)	(0.50)	(0.63)	(0.50)	(0.50)	(0.22)
Income: Irregular and Reduced	0.12	0.23	0.11**	0.08	0.25	0.17***	0.09	0.24	0.15***
	(0.32)	(0.42)	(0.00)	(0.27)	(0.44)	(0.00)	(0.29)	(0.43)	(0.00)
Income: No income sources	0.01	0.01	-0.00	0.02	0.02	-0.00	0.02	0.01	-0.00
	(0.12)	(0.11)	(0.94)	(0.14)	(0.13)	(0.85)	(0.13)	(0.12)	(0.71)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Source: rCARI survey 2023

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Summary and Key Points

- Compared to CARI, rCARI significantly reports lower number of food insecurity (by between 10 & 23 percentage points depending on the indicator)
  - 10 p.p. when remote FES is used
  - 23 p.p. when income indicator is used
- The income choice & income change variable generally tends to underreport AFI compared to FES
  - Driven by how small-scale farmers are categorized

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Summary and Key Points

- Regardless of the modality (F2F or Remote) is stable in the same direction
- Households tend to report higher, varied FC & higher FE
- The CATI may not be appropriate for areas/regions where with low phone penetration since it excludes most vulnerable households
- Households tend to report better FC and worse coping and livelihood situations (rCSI & LCS) over the phone

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Limitations of the study

- High attrition in the remote sample
- Responses of the sample that received phones could be influenced by the receipt of phones
- Low phone ownership in CAR (in our sample it was 25%, which is close to the national average of 23%)

Next steps …

- Further analysis to explore association between FES & income indicator
- Explore the re-categorization of the income sources into the 4-point rCARI income groups to align more with FES
- Continue the data collection to other pilot countries (Burkina Faso, Lebanon, Guatemala/Mauritania)

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Thank you

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Supplementary Slides

- Analysis restricted to only ADMIN5 Areas with at least 10 completed remote interviews
- Filter out ADMIN5 with no mobile phone ownership and or no respondents completing remote survey
- Restrict analysis to ADMIN5 areas with similar survey completion

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Additional Analysis [ADMIN5 Areas]

Table 3: Impact of Survey Modality: Standard CARI-FES vs. Standard rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.01	0.01	-0.00	0.00	0.01	0.01	0.00	0.01	0.01
	(0.09)	(0.09)	(0.92)	(0.00)	(0.12)	(0.32)	(0.06)	(0.10)	(0.41)
Marginally Food Secure	0.32	0.53	0.20***	0.13	0.16	0.04	0.21	0.40	0.19***
	(0.47)	(0.50)	(0.00)	(0.33)	(0.37)	(0.44)	(0.41)	(0.49)	(0.00)
Moderately Food Insecure	0.59	0.45	-0.15**	0.74	0.68	-0.05	0.67	0.53	-0.15***
	(0.49)	(0.50)	(0.01)	(0.44)	(0.47)	(0.37)	(0.47)	(0.50)	(0.00)
Severely Food Insecure	0.08	0.02	-0.06**	0.13	0.14	0.00	0.11	0.06	-0.05*
	(0.27)	(0.15)	(0.01)	(0.34)	(0.35)	(0.95)	(0.31)	(0.24)	(0.03)
Food Insecure Group	0.67	0.47	-0.20***	0.87	0.82	-0.05	0.78	0.59	-0.19***
	(0.47)	(0.50)	(0.00)	(0.33)	(0.39)	(0.30)	(0.41)	(0.49)	(0.00)
Observations	243	137	380	306	73	379	549	210	759

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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Additional Analysis [ADMIN5 Areas]

Table 4: Impact of Survey Modality: CARI-FES vs. rCARI-FES

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.01	0.01	-0.00	0.00	0.01	0.01	0.00	0.01	0.01
	(0.09)	(0.09)	(0.92)	(0.00)	(0.12)	(0.32)	(0.06)	(0.10)	(0.41)
Marginally Food Secure	0.32	0.29	-0.03	0.13	0.14	0.01	0.21	0.24	0.02
	(0.47)	(0.46)	(0.56)	(0.33)	(0.35)	(0.83)	(0.41)	(0.43)	(0.47)
Moderately Food Insecure	0.59	0.58	-0.01	0.74	0.62	-0.12	0.67	0.60	-0.08*
	(0.49)	(0.49)	(0.87)	(0.44)	(0.49)	(0.05)	(0.47)	(0.49)	(0.05)
Severely Food Insecure	0.08	0.12	0.04	0.13	0.23	0.10	0.11	0.16	0.05
	(0.27)	(0.32)	(0.24)	(0.34)	(0.43)	(0.06)	(0.31)	(0.36)	(0.09)
Food Insecure Group	0.67	0.70	0.03	0.87	0.85	-0.02	0.78	0.75	-0.03
	(0.47)	(0.46)	(0.55)	(0.33)	(0.36)	(0.61)	(0.41)	(0.43)	(0.37)
Observations	243	137	380	306	73	379	549	210	759

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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Additional Analysis [ADMIN5 Areas]

Table 5: Impact of Survey Modality: CARI-Inc vs. rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.02	0.01	-0.01	0.00	0.01	0.01	0.01	0.01	-0.00
	(0.14)	(0.09)	(0.26)	(0.06)	(0.12)	(0.46)	(0.10)	(0.10)	(0.86)
Marginally Food Secure	0.54	0.53	-0.01	0.35	0.16	-0.19***	0.44	0.40	-0.04
	(0.50)	(0.50)	(0.80)	(0.48)	(0.37)	(0.00)	(0.50)	(0.49)	(0.38)
Moderately Food Insecure	0.42	0.45	0.03	0.62	0.68	0.07	0.53	0.53	0.00
	(0.49)	(0.50)	(0.58)	(0.49)	(0.47)	(0.27)	(0.50)	(0.50)	(0.99)
Severely Food Insecure	0.02	0.02	-0.00	0.03	0.14	0.11**	0.03	0.06	0.04*
	(0.16)	(0.15)	(0.86)	(0.16)	(0.35)	(0.01)	(0.16)	(0.24)	(0.04)
Food Insecure Group	0.44	0.47	0.03	0.64	0.82	0.18***	0.55	0.59	0.04
	(0.50)	(0.50)	(0.62)	(0.48)	(0.39)	(0.00)	(0.50)	(0.49)	(0.36)
Observations	243	137	380	306	73	379	549	210	759

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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Additional Analysis [ADMIN5 Areas]

Table 6: Impact of Indicator Composition: CARI-FES vs. CARI-Inc in F2F Surveys (Paired)

	Phone Owners			No Phone			Full Sample		
Food Security Category	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff
Food secure	0.01	0.02	0.01	0.00	0.00	0.00	0.00	0.01	0.01*
	(0.01)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Marginally Food Secure	0.32	0.54	0.22***	0.13	0.35	0.23***	0.21	0.44	0.22***
	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
Moderately Food Insecure	0.59	0.42	-0.18***	0.74	0.62	-0.12**	0.67	0.53	-0.15
	(0.03)	(0.03)	(0.04)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
Severely Food Insecure	0.08	0.02	-0.05**	0.13	0.03	-0.11***	0.11	0.03	-0.08
	(0.02)	(0.01)	(0.02)	(0.02)	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)
Food Insecure Group	0.67	0.44	-0.23***	0.87	0.64	-0.23***	0.78	0.55	-0.23
	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Observations	243	243	243	306	306	306	549	549	549

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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Additional Analysis [ADMIN5 Areas]

Table 6: Impact of Indicator Composition: CARI-FES vs. CARI-Inc in F2F Surveys (Paired)

	Phone Owners			No Phone			Full Sample		
Food Security Category	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff
Food secure	0.01	0.02	0.01	0.00	0.00	0.00	0.00	0.01	0.01*
	(0.01)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Marginally Food Secure	0.32	0.54	0.22***	0.13	0.35	0.23***	0.21	0.44	0.22***
	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
Moderately Food Insecure	0.59	0.42	-0.18***	0.74	0.62	-0.12**	0.67	0.53	-0.15
	(0.03)	(0.03)	(0.04)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
Severely Food Insecure	0.08	0.02	-0.05**	0.13	0.03	-0.11***	0.11	0.03	-0.08
	(0.02)	(0.01)	(0.02)	(0.02)	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)
Food Insecure Group	0.67	0.44	-0.23***	0.87	0.64	-0.23***	0.78	0.55	-0.23
	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Observations	243	243	243	306	306	306	549	549	549

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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Additional Analysis [ADMIN5 Areas]

Table 7: Impact of Indicator Composition: CARI-FES vs. CARI-Inc in Remote Surveys (Paired)

	Phone Owners			No Phone			Full Sample		
Food Security Category	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff
Food secure	0.01	0.01	0.00	0.01	0.01	0.00	0.01	0.01	0.00
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)
Marginally Food Secure	0.29	0.53	0.23***	0.14	0.16	0.03	0.24	0.40	0.16***
	(0.04)	(0.04)	(0.05)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
Moderately Food Insecure	0.58	0.45	-0.14**	0.62	0.68	0.07	0.60	0.53	-0.07
	(0.04)	(0.04)	(0.06)	(0.06)	(0.05)	(0.06)	(0.03)	(0.03)	(0.04)
Severely Food Insecure	0.12	0.02	-0.09***	0.23	0.14	-0.10	0.16	0.06	-0.10
	(0.03)	(0.01)	(0.03)	(0.05)	(0.04)	(0.05)	(0.03)	(0.02)	(0.02)
Food Insecure Group	0.70	0.47	-0.23***	0.85	0.82	-0.03	0.75	0.59	-0.16***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)	(0.03)	(0.03)	(0.03)	(0.03)
Observations	137	137	137	73	73	73	210	210	210

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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Additional Analysis [ADMIN5 Areas]

Table 14: Logit Regression on Food Security based Outcomes

	Food Insecure with FES			Food Insecure with Income		
	(1)	(2)	(3)	(4)	(5)	(6)
Household owns phone	-0.19***	-0.14***	-0.14***	-0.25***	-0.17**	-0.17**
	(0.03)	(0.03)	(0.03)	(0.06)	(0.06)	(0.06)
Remote Modality	0.01	0.01	0.01	0.09	0.11	0.11
	(0.06)	(0.07)	(0.07)	(0.05)	(0.06)	(0.06)
List respondent age		-0.00	-0.00		0.00	0.00
		(0.00)	(0.00)		(0.00)	(0.00)
Household head is the respondent		0.02	0.03		-0.10*	-0.09*
		(0.04)	(0.04)		(0.05)	(0.05)
Female Household Head		0.04	0.04		0.03	0.04
		(0.04)	(0.04)		(0.05)	(0.05)
Married Household Head		-0.01	-0.01		-0.08	-0.08
		(0.05)	(0.05)		(0.05)	(0.05)
Household head years of education		-0.00	-0.00		-0.01	-0.01
		(0.00)	(0.00)		(0.01)	(0.01)
Household size		0.00	0.00		-0.01	-0.01
		(0.01)	(0.01)		(0.01)	(0.01)
Number of household member younger than 15		-0.02*	-0.02*		-0.02	-0.02
		(0.01)	(0.01)		(0.01)	(0.01)
Number of household member older than 60		0.03	0.03		0.05	0.05
		(0.04)	(0.04)		(0.05)	(0.06)
Has household member with disability/chronic disease		-0.08**	-0.08**		-0.06	-0.06
		(0.03)	(0.03)		(0.04)	(0.04)
Household Crowding Index		0.00	0.00		0.01	0.01
		(0.01)	(0.01)		(0.01)	(0.01)
Household has impoverished roof		0.05	0.05		-0.02	-0.02
		(0.03)	(0.03)		(0.07)	(0.07)
Household has impoverished floor		0.02	0.03		0.04	0.04
		(0.06)	(0.06)		(0.06)	(0.06)
Short Form			-0.03			-0.04
			(0.03)			(0.06)
Obs.	759	759	759	759	759	759

Note: Estimates are marginal effects using logit model regression. Robust standard errors are clustered at village level (Admin5) and reported in parentheses. * denotes statistical significance at 0.05 ** at 0.01 and *** at 0.001.

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